

08 · Did the loyalty rollout work? — difference-in-differences (CausalPy)

July 6, 2026

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The decision. We launched the loyalty app in some stores, not others. Revenue in the launch stores went up — but revenue went up *everywhere* (a seasonal tide). Did the app add anything **net of the tide**, and is it worth rolling out chain-wide?

Difference-in-differences subtracts two differences: (treated after — before) — (control after — before). The shared seasonal move cancels — *if* treated and control stores would have moved in parallel absent the app. We test that with an **event study**, run a **placebo**, estimate the effect with uncertainty, turn it into a €/store/week rollout case, and flag the **staggered-adoption trap** with a live demonstration.

7-step contract.

1.1 2 · Simulate a ground truth

40 stores, half get the loyalty app at week 12. Every store shares a seasonal pattern and has its own baseline; the app adds a **true €400/store/week**. Treated and control move in **parallel** before launch by construction — so DiD should recover €400, and the event study should show a flat pre-trend.

TRUE effect = €400/store/week · 40 stores, 24 weeks, launch week 12

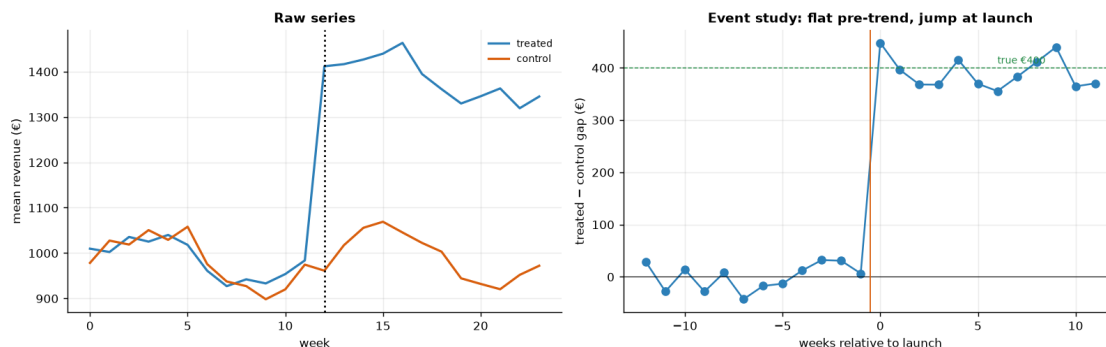
	store	unit	t	week	group	post_treatment	post	revenue
0	store_00	0	0	0	1	False	0	1071.613014
1	store_00	0	1	1	1	False	0	1066.597066
2	store_00	0	2	2	1	False	0	1019.187212
3	store_00	0	3	3	1	False	0	834.246887
4	store_00	0	4	4	1	False	0	1030.278745

1.2 3 · Identify — the 2x2 estimand, parallel trends, and the event study

DiD estimates β_3 in $Y = \beta_0 + \beta_1 \text{group} + \beta_2 \text{post} + \beta_3 (\text{group} \times \text{post}) + \varepsilon$, which equals the difference of the two before/after differences. **Key assumption — parallel trends:** absent the app, treated

and control revenue would have moved together. It's untestable post-launch, but the **event study** — the treated-minus-control gap in *each* period relative to launch — is the evidence: flat and ~0 before launch supports it; a jump at launch is the effect.

Pre-launch gaps average €22 (~0 supports parallel trends); post-launch gaps jump toward €400.



1.3 4 · Estimate — Bayesian difference-in-differences

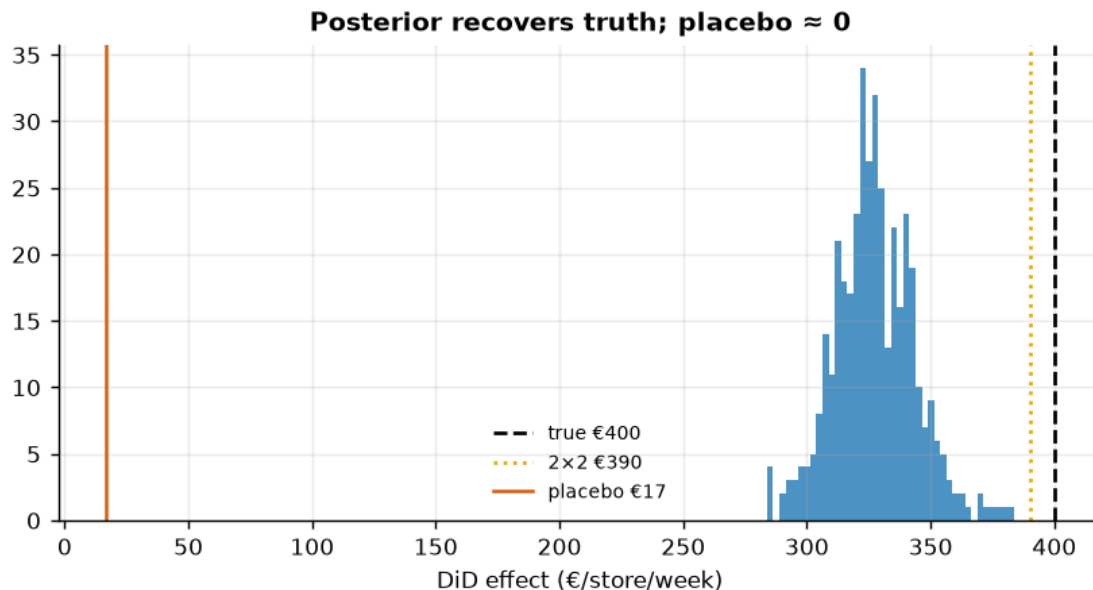
CausalPy's `DifferenceInDifferences` expects the classic **2x2 form** — one pre and one post observation per unit — so we collapse each store's weekly revenue to a pre- and post-launch mean.

DiD effect €327/store/week (true €400) · 90% CI [€302, €353]

1.4 5 · Validate — 2x2 cross-check and a placebo test

Cross-check the Bayesian estimate against the hand-computed 2x2 (the priors mildly shrink it), then **falsify**: run DiD on the *pre-period only*, splitting it into a fake “before/after” at week 6. There was no treatment then, so the placebo effect should be ~ 0 — a large one would mean our design manufactures effects.

Bayesian €327 · 2x2 €390 · true €400 · placebo (no real effect) €17



1.5 6 · Decide, in euros — the rollout case

Net €207/store/week · projected annual value €5,389,807 [90% €4,728,804, €6,069,367]

P(app pays) 1.00 → ROLL OUT

Break-even: the app stays profitable up to a running cost of ~€302/store/week (5th pct of the effect).

1.6 7 · Caveats — and the staggered-adoption trap, demonstrated

Parallel trends is the whole ballgame and untestable post-launch; a diverging pre-trend kills the design (use synthetic control, nb 07). **No spillover** between nearby stores. But the subtlest trap is **staggered adoption**: when units adopt at *different* times, naive two-way fixed-effects DiD uses already-treated units as controls (“forbidden comparisons”) and can put **negative weights** on some effects (Goodman-Bacon), biasing the estimate even when every true effect is positive. We show it live:

Naive TWFE €177 vs true average post-treatment effect €400 - bias -223. Use Callaway-Sant'Anna / Sun-Abraham for staggered rollouts.

